

Group 1
Samples

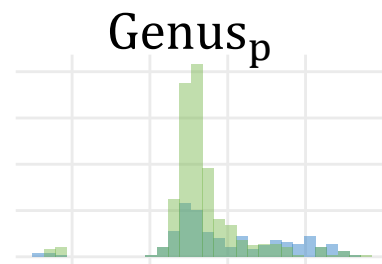
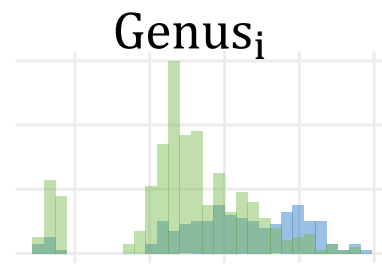
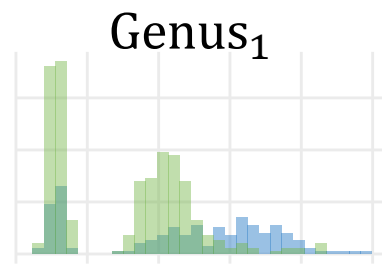
CLR transformed matrix

		Genera				
		1	...	i	...	p
1		5.139	...	1.393	...	2.374
$\vdots$		$\vdots$	...	$\vdots$	...	$\vdots$
k		-1.395	...	3.128	...	1.806
$\vdots$		$\vdots$	...	$\vdots$	...	$\vdots$
n_1		4.983	...	-0.506	...	-0.913

Group 2
Samples

n_1+1		2.278	...	3.847	...	-1.349
$\vdots$		$\vdots$	...	$\vdots$	...	$\vdots$
l		-1.721	...	2.491	...	1.519
$\vdots$		$\vdots$	...	$\vdots$	...	$\vdots$
n		3.841	...	1.896	...	2.301

Visualization of distributions per group



CLR coordinate

■ Group 1 ■ Group 2

Two-stage testing approach (one test per genus)

Hypotheses:

H_0 : No group difference in distribution of genus i

H_1 : Group difference in distribution of genus i

Testing:

Two stage approach with the *waddR* package

→ Separate test statistic for zero and non-zero part

→ Three p -values per genus:

- Zero part
- Non-zero part
- Combined p -value